DOINGG@UNION.EDU

DOING-LAB-NOTEBOOK

SELF

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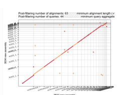 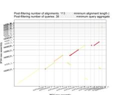 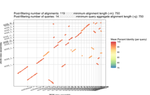

The shared electronic lab notebook of the Doing Com-Bio Lab.

Announcements

Something

Something else

Schedule

Weekly Schedule

Schedule

check often as things may change

W	TOPIC	Reading	Due
1	What is bioinformatics: what are computers and living cells good at?	Bioinformatics for Dummies (BFD) Ch. 1 & 2	Survey, Quiz 0
2	Algorithms & the Genetic Code	Think Python (TP) Ch. 1, 2, 3.4–3.7 & Molecular & Cell Biology (MCFD) Ch. 1, 6, 7	HW 1, Quiz 1
3	Modular Programming & Proteins	TP Ch. 5.1–5.7, 5.11, 6.1 & MCFD Ch. 16, 17	HW 2, Quiz 2
4	Loops & Sequencing	TP Ch. 6.1–6.1, 7.1–7.3, 7.7 & MCFD Ch. 19	HW 3, Quiz 3
5	Sequence Processing & Disease	BFD Ch. 7	HW 4, Quiz 4
6	Sequence Alignment	BFD Ch. 9, TP Ch. 8	HW 5, Quiz 5
7	Multiple Sequence Alignment	BFD Ch. 13	Project 1 Outline

W	TOPIC	Reading	Due
8	Polymerase Chain Reaction	BDF Ch. 13, 9 cont.	Project 1
9	Phylogenetics	BFD Ch. 13 cont.	Project 2 Check-in
10	Gene Prediction	BFD Ch. 5, pp. 145-153	Project 2

Meeting Notes

Lab Meeting 1

Lab Meeting: 1

June 15, 2026

1. lab meeting structure (first few weeks)
   - 10–15 mins each for project Q&A
   - maybe walk through notebooks from the previous week, if helpful
   - flash journal club
2. daily scrum – slack
   - home or lab
   - what you'll focus on
     - to know what Qs to expect or work together
     - to avoid doing the same thing
     - to get ideas from each other
   - approx hours
     - to synchronize if helpful
     - to know when to expect Qs or As
3. working notebooks vs. notebook notebooks
   - Medium post on good jupyter nb habits
   - working colab for now – for reproducibility
   - data stored in the same google drive folder should be reproducible across accounts..
   - Union supposed to be getting a compute cluster
   - notebook notebooks can be edited in anything (I use VS code)
     - don't require reading in data?

4. github seqADAGE https://github.com/georgiadoing/seqADAGE
   - fork
   - clone
   - make project folder within `/Py/`
   - we'll start code review in a couple weeks
   - github tutorials
     - maybe try the first..
5. data in google drive folder (for now)
   - later will use Open Science Framework: https://osf.io/
   - maybe this one but maybe a *new* one
   - could create accounts (union email?)
6. Misc.
   - my office key and keycard

Lab Meeting 2

Lab Meeting: 2

June 22, 2026

Weekly Notes

GD Week 1

GD Weekly Notes: 1

June 15, 2026

Shared/Overall To-do

- ☐ Implement tuner method in adage class
- ☐ E coli gene lists for enrichment analyses
- ☒ Get lab notebook setup
- ☐ Decide on repo structure
- ☐ Plan sampling excursion
- ☒ Key access to CAP/crochet
- ☒ Schedule lab meetings - update calendar
- ☒ Group meeting once a week
- ☐ 1-1 meetings also weekly or as-needed

Phylo-learning To-do

- ☒ Get DNA sequences into google drive
- ☐ Get P.a. DNA seq
- ☐ S aureus gene lists for enrichment analysis
- ☐ P aeruginosa gene lists for enrichment analysis

Bio-xAI To-do

- ☐ model to test SHAP tutorial on

- [ ] colaboration with Josh/Ara

About

References